

1 Introduction

In guitar, tone control is a major technical skill. Each musical instrument’s quality of sound—its timbre—is differentiated by the varying composition of harmonic frequencies f_n , which are present in addition to the fundamental frequency f_1 (Tsokos, 2014). On top of fundamental variations between instruments, each instrument’s timbre also varies based on how notes are played, like the position of the picking plectrum along the string, the force applied, and the angle of picking (Perov et al., 2016). It is qualitatively observed that plucking near the bridge will create a harsher, “tinny” sound while plucking toward the bridge will create a fuller sound.

This investigation will explore how the quality of sound as measured by the output signal may change depending on where the string is plucked. The research question I have decided to investigate is: *How does changing the position of the plectrum along a guitar’s fretboard affect the frequency of the harmonic with the highest intensity when a string with constant tension is plucked?*

1.1 Background

1.1.1 Standing Waves

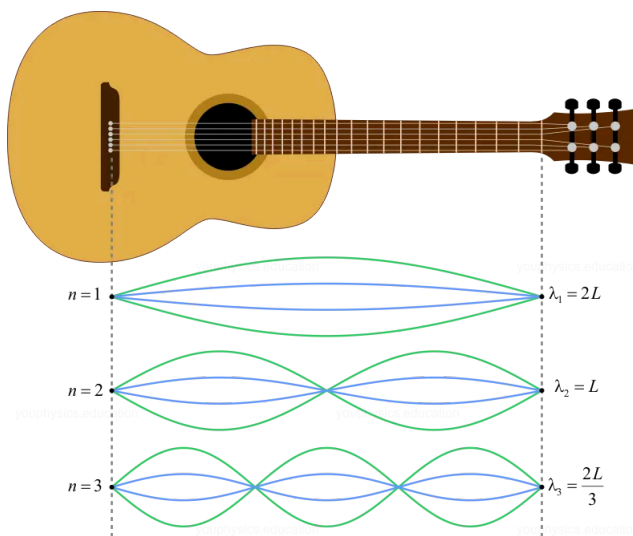


Figure 1: Standing waves on a guitar string (Novak, n.d.)

A guitar string is a string with two fixed ends. When the string is plucked, a transverse wave is initiated that propagates along the length of the string. The reflection of the transverse wave at the fixed ends superimposes with the original wave, forming nodes (points of zero displacement) and antinodes (points of maximum displacement). This creates a standing wave, which is a wave that does not propagate but rather oscillates in place (Tsokos, 2014).

The first harmonic is a standing wave with a single loop made of two nodes and one antinode, which results in a sound wave at the fundamental frequency f_1 , given by the equation $f_1 = \frac{v}{2L}$ where v is the wave speed and L is the length of the string.

The linear principle of superposition states that the net displacement of a system subjected to multiple independent disturbances (waves, in this case) is the algebraic sum of the displacements due to each disturbance. If $y_1(x, t)$ and $y_2(x, t)$ are two waveforms propagating along the same string, then the net waveform $y(x, t)$ is given by the equation $y(x, t) = y_1(x, t) + y_2(x, t)$.

Due to this principle, multiple, independent standing waves are formed on the guitar string, which results in higher-order harmonics. Each subsequent harmonic has a frequency that is an integer multiple of the fundamental frequency, given by

$$f_n = n f_1 \quad (1)$$

where n is the harmonic number. This is known as frequency quantization.

These individual waveforms superimpose linearly to form a composite waveform which is the sum of all individual harmonic waveforms. As a result, the sound emitted by a guitar string is a mixture of multiple harmonic frequencies, as in Figure 1.

1.1.2 Theoretical Model

To investigate how the picking length affects the harmonic with the highest amplitude, we must determine the variation of the amplitude of each harmonic frequency separately.

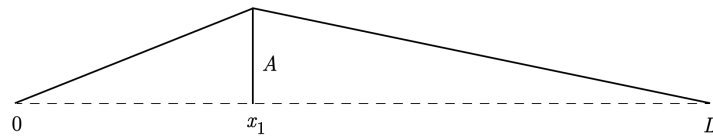


Figure 2: A guitar string of length L pulled a distance A at position x_1 (Perov et al., 2016)

The amplitude b_n of the n th harmonic is given by the equation

$$b_n = \frac{2A \sin\left(n\pi \frac{x_1}{L}\right)}{\frac{x_1}{L} \left(1 - \frac{x_1}{L}\right) \pi^2 n^2} \quad (2)$$

where:

- x_1 is the picking position/distance from the nut
- A is the distance the string is pulled when plucked
- L is the length of the string (Perov et al., 2016)

The variation in amplitude thus depends on the harmonic n , like so:

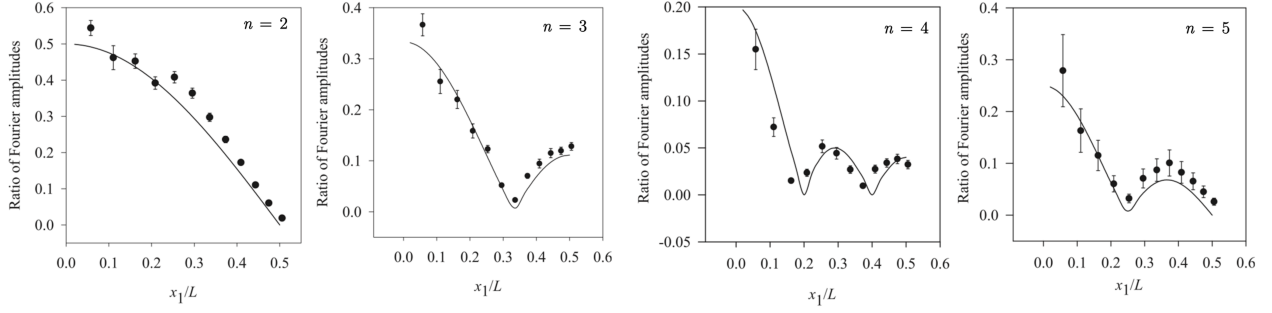


Figure 3: Variation of the amplitude b_n for $n = 2$ through $n = 5$ with the picking length $\frac{x_1}{L}$ (ratio of picking position to string length L) (Perov et al., 2016)

By calculating b_n for $n = 2 \dots 10$ at each variation of x_1 and choosing the value of n that maximizes b_n , the non-fundamental harmonic frequency with the highest amplitude can be calculated using Equation 1. The fundamental frequency f_1 of the high E string (see Section 2.1) is taken to be $f_1 = 329.63$ Hz.

Plotting the picking length x_1 against the non-fundamental harmonic frequency with the highest amplitude f results in a Gaussian-like function where the 2nd harmonic has the highest amplitude across picking lengths, except near the middle of the fretboard where the 3rd harmonic has the highest amplitude.

This is shown in Figure 4, with a Gaussian trendline fitted to the data points, in the form

$$y = A \times e^{-\frac{(x-B)^2}{C^2}} + D \quad (3)$$

Fitting the trendline to the data points results in the following parameters:

- $A = 329.63$ representing the amplitude of the peak above the baseline. This is equal to the fundamental frequency f_1 because the peak (3rd harmonic) is one harmonic above the baseline (2nd harmonic). Recall that each subsequent harmonic frequency is an integer multiple of the fundamental frequency.
- $B = 0.29$ representing the picking length where the peak occurs.
- $C = 0.026$ representing the width of the peak (as chosen by the curve fitting algorithm).
- $D = 659.26$ representing the baseline frequency, which is the frequency of the 2nd harmonic.

resulting in the specific trendline

$$y = 329.63 \times e^{-\frac{(x-0.29)^2}{0.026^2}} + 659.26 \quad (4)$$

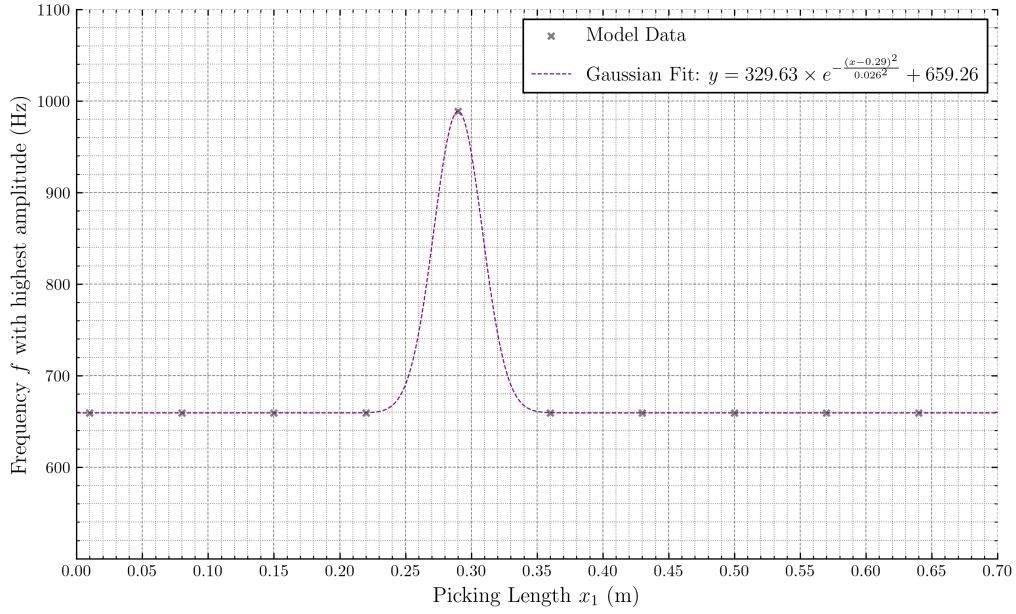


Figure 4: Theoretical Gaussian trend of the variation of the non-fundamental harmonic frequency with the highest amplitude across picking lengths x_1

Therefore, it is hypothesized that the non-fundamental harmonic frequency with the highest amplitude will remain constant across picking lengths x_1 , except with a peak around the middle of the fretboard ($x_1 = \frac{L}{2}$).

2 Methodology

2.1 Variables

Independent Variable

Picking length x_1 (m), the distance in meters from the nut where the string is plucked/disturbed

- $x_1 = \{0.01\text{m}, 0.08\text{m}, 0.15\text{m}, 0.22\text{m}, 0.29\text{m}, 0.36\text{m}, 0.43\text{m}, 0.50\text{m}, 0.57\text{m}, 0.64\text{m}\}$
- The variations are chosen to be evenly spaced across the guitar's fretboard/scale length (the distance between the nut and the bridge), which is 0.64m. This will allow the RQ to be investigated across the entire fretboard, and mirrors the methodology of Perov et al. (2016).
- The IV is varied by plucking at the picking length as indicated by a tape measure.

Dependent Variable

Frequency f of the non-fundamental harmonic with the highest amplitude (Hz)

- Measured by recording the frequency f_n of the n th harmonic that has the highest amplitude through FFT (Fast Fourier Transform) analysis, where $n = 2 \dots 10$.
- This DV was chosen because the highest intensity harmonic—the harmonic that is most audible—has a significant impact on the quality/timbre of sound.

Controlled Variables

Control	Reason	Specification
Diameter, tension, age, and other material properties of the string	Different material properties of the string may change its harmonic composition.	The same high E string tuned to the standard $E = 329.63$ Hz will be used for all the experiments. This will also control the density of the string, as data collection is performed in the same room with ambient temperature 23°C .
Guitar body and electronics	To minimize random error from the frequency response imparted by the pickups and guitar body.	The neck pickup setting on the same guitar/pickup combination will be used across trials.
Plectrum choice and picking technique	Different plectrums and picking techniques may affect the harmonic composition. This also controls A in Equation 2.	The same pick will be used across trials. The string will be plucked at a 45° angle to the fretboard, with the tip of the pick contacting the fretboard.

Table 1: Controlled Variables

2.2 Apparatus & Experimental Setup

- Electric guitar (6-string “Kiesel Guitars Osiris” used) with 0.009mm gauge nylon electric guitar string (“D’Addario XS Nickel XSE0942 Super Light Coated Electric Guitar Strings” used)
- 1 audio interface with instrument input (*Audient EVO4* used)
- 1 instrument cable
- 1 Computer with audio analysis and spreadsheet software (*MATLAB* and *Google Sheets* used)
- 1 tape measure
- 1 clamp stand
- Masking tape

Figure 5 presents the experimental setup used to collect data. The guitar is clamped down for ease of plucking, and the tape measure is placed directly on the 2nd and 3rd strings to be on the same plane as the 1st string (which is plucked).



Figure 5: Labeled photograph of experimental setup

2.3 Experimental Procedure

PREPARATION

1. Clamp the guitar's head using the clamp stand to hold it in place.
2. Lay the tape measure out along the fretboard on top of the 2nd and 3rd strings, securing it in place with the clamp stand and masking tape.
3. Connect the guitar to the audio interface using the instrument cable, and connect the audio interface to the computer.
4. Switch to the neck pickup on the pickup selector (most inward setting), turn all the knobs on the guitar to 10 (maximum position clockwise), and adjust the gain on the audio interface appropriately to prevent clipping of the audio signal.

RECORDING AND ANALYSIS

5. Start the 5-second recording on the computer.
6. At the first variation of x_1 , pluck the 1st string using the pick at a 45° angle to the fretboard, touching the pick to the fretboard and pushing towards the string perpendicularly.
7. Wait for the recording period to end and ensure the data is saved.
8. Using the frequency analysis software, convert the time-domain data into frequency-domain data.
9. Identify the true fundamental frequency by searching for a peak in the data within 5% of the expected fundamental frequency $f_1 = 329.63$ Hz
10. Calculate the frequency of each non-fundamental harmonic n for $n = 2 \dots 10$
11. Calculate the absolute amplitude $|b_n|$ for the frequency at each harmonic

12. Record the harmonic n that maximizes the absolute amplitude $|b_n|$ and calculate its frequency f_n .
13. Repeat steps 5–12 four more times to complete five trials for the variation.
14. Repeat steps 5–13 for the remaining variations, modifying the picking length x_1 according to Section 2.1.

Note: In data collection for the experiment, steps 8-12 were automated using a custom MATLAB program.

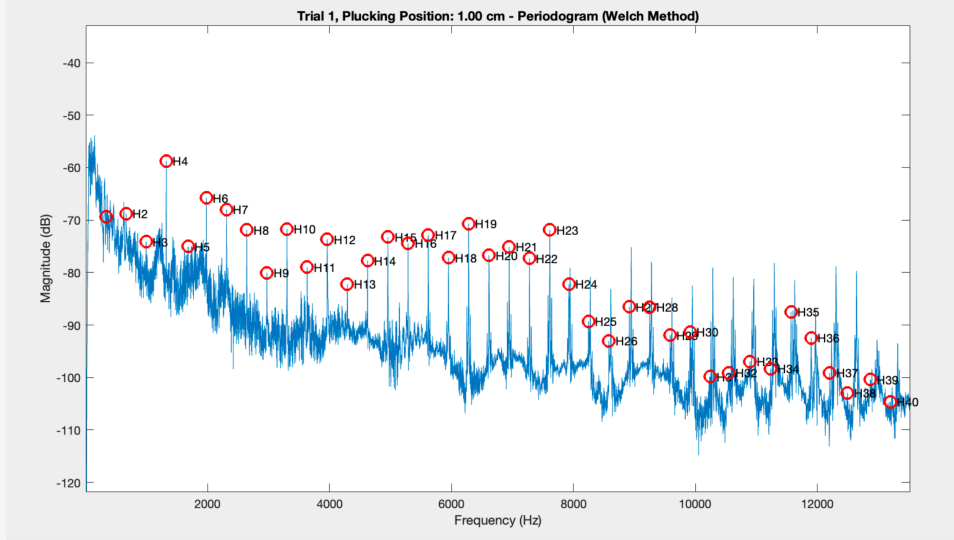


Figure 6: Example of frequency-domain data showing harmonic peaks

As an example, Figure 6 shows an example of the resulting frequency-domain data (step 8), which has been converted by the program from the time series data using FFT (omitted because it cannot be represented meaningfully in text/table form). The FFT process involves applying a Hann window to the recorded audio data to reduce noise interference and computing the Fourier transform of the windowed data using Welch’s method (MathWorks, 2023).

The fundamental is identified by looking for peaks around the expected frequency (step 9), allowing for variations in tuning. This window is set to 5% of the standard tuning value $f_1 = 329.63\text{Hz}$. The harmonic frequencies are then calculated according to Equation 1 (step 10). For illustration, each harmonic frequency is represented by a peak in the data and is marked $\bigcirc$ H2 (2nd harmonic) through $\bigcirc$ H40 (40th harmonic) for clarity.

2.4 Safety, Ethical, and Environmental Considerations

This is a safe experiment. As the guitar is directly connected to the audio interface, any voltages are low and are not likely to pose a safety risk. Water is not used therefore water damage is not a risk. New strings are not required and no consumable materials apart from masking tape are used, limiting the environmental impact. There are no significant ethical considerations.

3 Analysis

3.1 Raw Data - Frequency-Domain

Plucking length x_1 (m) $\Delta x_1 = \pm 0.005\text{m}$	Frequency with highest amplitude $f / \Delta f$ (Hz)					
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Δf
0.010	655.70	655.88	655.88	655.88	655.88	32.96
0.080	655.70	655.88	655.88	655.88	655.88	32.96
0.150	655.88	655.88	655.88	656.07	655.88	32.96
0.220	656.07	655.88	655.88	656.07	656.07	32.96
0.290	984.01	983.83	984.01	984.01	984.01	49.44
0.360	656.07	984.01 ¹	655.88	655.88	655.88	32.96
0.430	655.88	656.07	655.88	655.88	655.88	32.96
0.500	655.88	655.88	655.88	655.88	655.88	32.96
0.570	656.07	655.88	655.88	655.88	655.88	32.96
0.640	655.88	656.07	656.07	655.88	656.07	32.96

Table 2: Raw data of highest amplitude harmonic frequencies at each plucking length

QUALITATIVE OBSERVATIONS

- It was observed that plucking near the ends of the string created a harsher, “tinny” sound. On the other hand, plucking near the middle of the string created a fuller sound.
- This contrast is likely a result of a change in harmonic composition, where the harsher sound is a result of a reduced presence of mid frequencies, around 500Hz to 2000Hz, and the fuller sound is a result of a more balanced frequency spectrum.

¹Outlier removed for data processing

ABSOLUTE UNCERTAINTIES

- The absolute uncertainty in the picking length x_1 is $\Delta x_1 = \pm 0.005m$ as the tape measure has a precision of 1 cm.
- The absolute uncertainty Δf in the frequency f is

$$\Delta f = \pm n \times 0.05 f_1 = \pm n \times 16.48 \text{Hz} \quad (5)$$

as the frequency search window in the FFT is 5% of the expected fundamental frequency $f_1 = 329.63 \text{Hz}$. 5% is multiplied by n because the uncertainty propagates when the non-fundamental harmonic is calculated using Equation 1, which multiplies the fundamental frequency by the harmonic number n .

For instance, plucking length $x_1 = 0.01m$ results in the 2nd harmonic with the highest amplitude, so regardless of the FFT-found fundamental frequency, the absolute uncertainty is $\Delta f = \pm 2 \times 16.48 = \pm 32.96 \text{Hz}$.

3.2 Processed Data

Plucking length x_1 (m) $\Delta x_1 = \pm 0.005m$	Average frequency with highest amplitude $\bar{f}$ (Hz)	Uncertainty in $\bar{f}$ Δf (Hz)
0.01	655.85	32.96
0.08	655.85	32.96
0.15	655.92	32.96
0.22	655.99	32.96
0.29	983.97	49.44
0.36	655.93	32.96
0.43	655.92	32.96
0.50	655.88	32.96
0.57	655.92	32.96
0.64	655.99	32.96

Table 3: Processed data of average highest amplitude harmonic frequencies at each plucking length

CALCULATIONS

- The average frequency with the highest amplitude $\bar{f}$ was calculated by averaging the frequencies with the highest amplitude across trials, for each variation of the picking length x_1 . For example, for $x_1 = 0.01m$, the average frequency $\bar{f}$, rounded to match the original accuracy, is

$$\begin{aligned}\bar{f} &= \frac{1}{5} \sum_{n=0}^5 (f_{x_1=0.01, \text{ trial } n}) \\ &= \frac{1}{5} (655.70 + 655.88 + 655.88 + 655.88 + 655.88) \\ &= 655.85 \text{ Hz}\end{aligned}\tag{6}$$

- The uncertainties in frequency are due to systematic errors from the FFT process rather than random error between trials, therefore the absolute uncertainties are used as in Table 2 as opposed to $\Delta\bar{f} = \frac{f_{\max} - f_{\min}}{2}$. For instance, the absolute uncertainty for picking length $x_1 = 0.01m$ is 32.96Hz, so $\Delta\bar{f} = 32.96\text{Hz}$.

3.3 Graphical Analysis

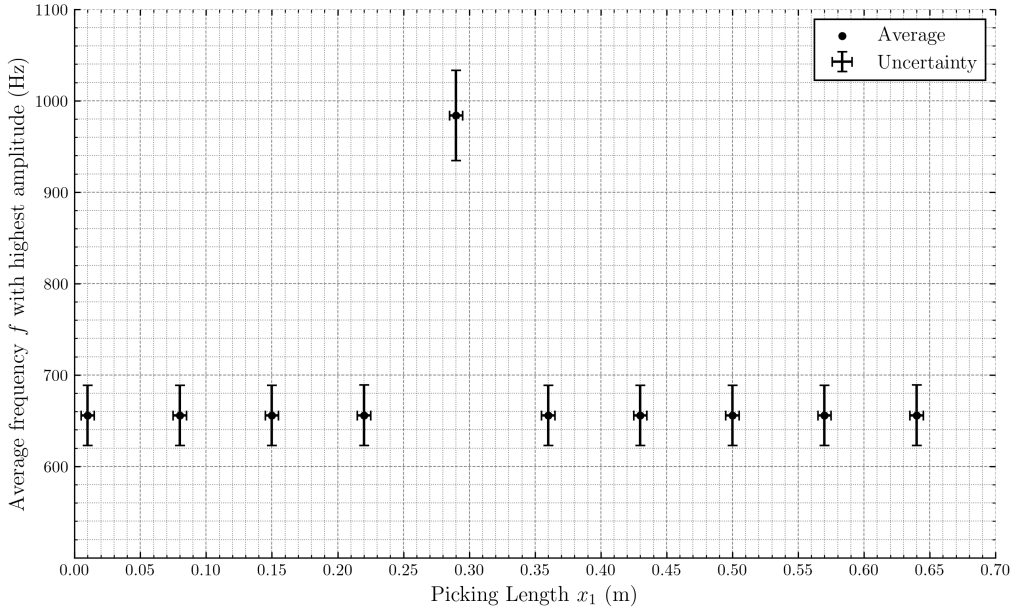


Figure 7: Graph of picking length x_1 against average frequency $\bar{f}$

Graphing the processed data, there is a clear peak at the data point (0.29, 983.97), while every other data point has a y-value around 655Hz $\pm$ 1Hz. The error bars are quite large, around ± 30 –50Hz, however, this does not impair the clear peak at 0.29m.

The graph cannot be linearized within the scope of the IA due to requiring Taylor series approximations (Ratkowsky & Dolby, 1975). Therefore, the data will be compared to the theoretical simulation data.

3.4 Comparison with Theoretical Data

As outlined in Section 1.1.2, the theoretical harmonic frequency with the highest amplitude f_{modeled} was calculated for each variation of the picking length x_1 . The collected data $\bar{f}_{\text{actual}}$ from the experiment (Section 3.2) was then compared with the theoretical data, as shown in Table 4 and Figure 8.

Plucking length x_1 (m) $\Delta x_1 = \pm 0.005\text{m}$	Measured average frequency with highest amplitude $\bar{f}_{\text{actual}}$ (Hz)	Theoretical frequency with highest intensity f_{modeled} (Hz)	Percentage difference between $\bar{f}_{\text{actual}}$ and f_{modeled} $\Delta\%$
0.01	655.85	659.26	-0.52%
0.08	655.85	659.26	-0.52%
0.15	655.92	659.26	-0.51%
0.22	655.99	659.26	-0.50%
0.29	983.97	988.89	-0.50%
0.36	655.93	659.26	-0.51%
0.43	655.92	659.26	-0.51%
0.50	655.88	659.26	-0.51%
0.57	655.92	659.26	-0.51%
0.64	655.99	659.26	-0.50%

Table 4: Comparison between $\bar{f}_{\text{actual}}$ and f_{modeled}

Overall, there is a low average percentage difference between the average frequency from the experiment and the theoretical value. The average percentage difference is -0.51%, an acceptable error, and is within the 5% margin of error from the FFT process (MathWorks, 2023).

This percentage difference is roughly constant across variations with the measured average frequency being around 3-5Hz lower than the theoretical, suggesting that there is a source of systematic error. This can likely be attributed to discrepancies in the tuning of the guitar string.

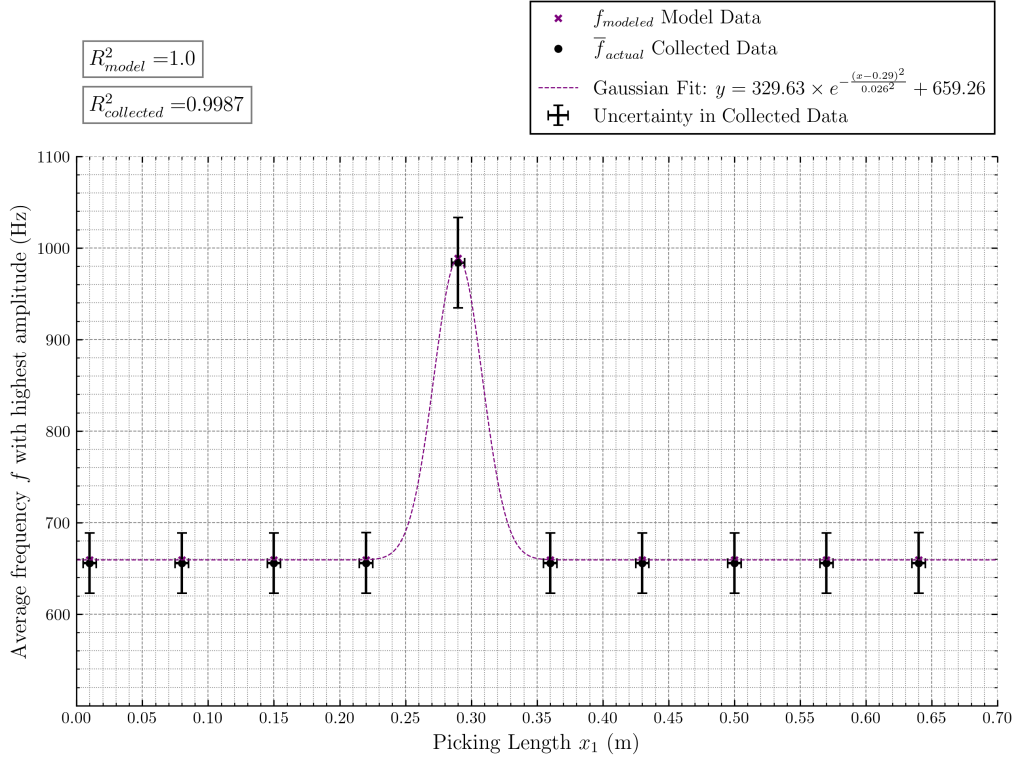


Figure 8: Comparison between $\bar{f}_{\text{actual}}$ and f_{modeled} with Gaussian trendline

Shown in Figure 8, plotting the theoretical Gaussian trendline (Section 1.1.2, Equation 4) on top of the experimental data results in a coefficient of determination of $R^2_{\text{collected}} = 0.9987$, indicating a strong “goodness of fit” between the trendline and the data.

The coefficient of determination is a measure of how well the observed outcomes are replicated by the model, based on the proportion of total variation of outcomes explained by the model. An R^2 of 0.9987 means that the trendline accounts for 99.87% of the variance in the experimental data, indicating an exceptionally high level of fit between the trendline and the data points.

4 Conclusion

The hypothesis, which stated that *the non-fundamental harmonic frequency with the highest amplitude will remain constant across picking lengths x_1 , except with a peak around the middle of the fretboard*, is well-supported. There is a clear trend of constant frequency, with a peak at the middle of length L which is at $x_1 = \frac{L}{2} = 0.29m$.

Notably, the error bars are quite large, with the lowest being $\pm 32.96\text{Hz}$ and the highest being $\pm 49.44\text{Hz}$

In comparison with the theoretical data, to which a Gaussian fit has been applied, the peak at $(0.29, 983.97)$ aligns, and so do the other values. This theoretical trendline passes through every

error bar of the collected data, with an R^2 value of 0.9987 which indicates a strong correlation. This strongly supports the validity of the data collected.

4.1 Scientific Reasoning

The results, both measured and theoretical, are supported by the scientific background outlined in Section 1.1. The results show that there is a variation of higher (non-fundamental) harmonics at each picking length. For instance, at $x_1 = 0.01m$, the highest amplitude harmonic frequency is 655.85Hz on average, which is the 2nd harmonic. The frequency is thus roughly $2 \times f_1 = 2 \times 329.63\text{Hz}$, showing that the harmonic is an integer multiple of the fundamental.

The peak at (0.29, 983.97), which is the 3rd harmonic frequency, may be explained by an antinode on the 3rd harmonic loop existing at that location. When the string is plucked at its midpoint, the central antinode of the third harmonic is excited to its maximum extent; there will be maximum displacement where the third harmonic has its maximum amplitude (central antinode). Compared with the 2nd harmonic, which has a node at the midpoint, the 3rd harmonic might be excited more because the 2nd harmonic's node is a position of zero displacement.

5 Evaluation

5.1 Strengths

Overall, the data was precise and accurate. This is supported by the data in that there was only one outlier (colored in red in Table 2), which was removed before data processing. The average standard deviation between trials was 0.08Hz, a value well below the degrees of uncertainty, which indicates that the data was extremely precise. Comparing the collected data to the theoretical data led to a low average percentage difference of -0.51% and high coefficient of determination ($R^2 = 0.9987$) with the theoretical trendline, further indicating that the data was accurate.

The methodology allowed 5 trials of data to be collected for 10 variations of the IV, which resulted in a relevant investigation. The control variables chosen were relevant and limited random/systematic error. The method of investigation, FFT analysis, was appropriate for investigating the research question as it yielded more precise data than if other less discriminatory harmonic frequency detection methods were used, such as a phone frequency detector application or a guitar tuner. The use of an automatic search algorithm to identify the frequencies as opposed to manual identification also increased accuracy, leading to less systematic and random error.

5.2 Limitations

Source/effects of error	Significance of error	Possible improvements
Errors in the FFT analysis, such as the frequency search window and the resolution of the FFT (not accounted for). These uncertainties propagate to the calculated harmonic frequencies, resulting in both random and systematic errors in the calculated frequencies. This can be seen in the offset (systematic error) between the measured data points and the theoretical.	Medium significance. As 5 trials are performed for each variation, the random error is reduced, and the low standard deviation in the data suggests that this is only a minor source of error. However, the systematic error (shown by the $-3-5$ Hz offset) is more significant.	The frequency search window can be reduced to reduce the random error. The resolution of the FFT can be increased to reduce the systematic error.
The string was plucked by hand, which introduces random error in the picking technique. This could result in a variation in the harmonic composition of the string, introducing random or systematic error.	Medium significance. The high precision (indicated by the low standard deviation of 0.08Hz) in the data suggested that random error was low, but this could have been masked by frequency quantization.	Repeat the experiment, using a robot/mechanical plucking device to pluck the string with a known, consistent force.
The theoretical value may not be accurate due to assumptions made, such as assuming A (the distance the string is pulled when plucking) to be 1. This could have introduced systematic error.	Low significance. The theoretical value is only used to compare with the collected data and the low average percentage difference suggest that this is not a significant source of error.	Controlling A in experimental data collection would forego this assumption, and could be achieved by using a mechanical plucking device as previously described.

Table 5: Limitations of the design of the experiment

5.3 Further Study

To further explore the factors that affect a guitar's tone or harmonic composition, other IVs may be explored:

1	The experiment could be repeated with different types of guitar strings, such as nylon, copper, or steel strings, to see how each string material affects the harmonic composition of the sound (the non-fundamental harmonic frequency with the highest amplitude).		
	IV	material/type of string	DV the non-fundamental harmonic frequency with the highest amplitude
	Because different materials have varying densities, tensile strengths, and elasticities, it can be expected that: • Nylon strings, which have lower density and tension, will produce a warmer and mellower sound resulting in lower frequency harmonics being more prominent. • Copper-wound strings, which have higher density and tension, will produce a brighter and richer sound resulting in higher frequency harmonics being more prominent. • Steel strings, which have the highest density and tension, will produce a bright and crisp sound resulting in even higher frequency harmonics being more prominent.		
2	The experiment could be repeated with different types of guitar pickups, such as single-coil and humbucker pickups, to see the effect of each pickup's frequency response, and how it affects the harmonic composition of the sound.		
	IV	type of pickup (single-coil, humbucker, mix, active/passive pickups, etc.)	DV the non-fundamental harmonic frequency with the highest amplitude
	Because pickups impart a frequency response on the output signal, it can be expected that: • Single-coil pickups, which tend to accentuate higher frequencies, might result in the amplification of higher frequency non-fundamental harmonics. • Humbucker pickups, which are designed to reduce hum and noise, might result in a lower frequency non-fundamental harmonic being the most prominent as humbuckers could dampen the higher frequency harmonics to some extent.		

Table 6: Possible extensions to the investigation

References

- MathWorks. (2023, September 1). *Welch's Power Spectral Density Estimate - MATLAB Pwelch*.
<https://www.mathworks.com/help/signal/ref/pwelch.html>
- Novak, J. P. *Standing Waves. What Is a Harmonic?*. Retrieved November 19, 2023, from <https://www.youphysics.education/standing-waves/harmonics/>
- Perov, P., Johnson, W., & Perova-Mello, N. (2016). The Physics of Guitar String Vibrations. *American Journal of Physics*, 1, 38–43. <https://doi.org/10.1119/1.4935088>
- Ratkowsky, D. A., & Dolby, G. R. (1975). Taylor Series Linearization and Scoring for Parameters in Nonlinear Regression. *Applied Statistics*, 1, 109. <https://doi.org/10.2307/2346709>
- Tsokos, K. A. (2014). *Physics for the IB Diploma* (Sixth edition). Cambridge University Press.